

SOCIAL MEDIA ANALYTICS

Course Outcomes:

Understand the **concept of Social media**

Understand the **concept of social media Analytics and its significance.**

Learners will be able to **analyze the effectiveness of social media**

Learners will be able to **use different Social media analytics tools effectively and efficiently.**

Learners will be able to **use different effective Visualization techniques to represent social media analytics.**

Acquire the fundamental **perspectives and hands-on skills needed to work with social media data.**

Chapter 1

Social Media Analytics: An Overview

Core Characteristics of Social Media, Types of Social Media, Social media landscape, Need for Social Media Analytics (SMA), SMA in small & large organizations.

Purpose of Social Media Analytics, Social Media vs. Traditional Business Analytics, Seven Layers of Social Media Analytics, Types of Social Media Analytics, Social Media Analytics Cycle, Challenges to Social Media Analytics, Social Media Analytics Tools

Chapter 2

Social Network Structure, Measures & Visualization

Basics of Social Network Structure - Nodes, Edges & Tie

Describing the Networks Measures - Degree Distribution, Density, Connectivity, Centralization, Tie Strength & Trust

Network Visualization - Graph Layout, Visualizing Network features, Scale Issues.

Social Media Network Analytics - Common Network Terms, Common Social Media Network Types, Types of Networks, Common Network Terminologies, Network Analytics Tools.

Chapter 3

Social Media Text, Action & Hyperlink Analytics

Social Media Text Analytics - Types of Social Media Text, Purpose of Text Analytics, Steps in Text Analytics, Social Media Text Analysis Tools

Social Media Action Analytics - What Is Actions Analytics?
Common Social Media Actions, Actions Analytics Tools

Social Media HyperLink Analytics - Types of Hyperlinks, Types of Hyperlink Analytics, Hyperlink Analytics Tools

Chapter 4

Social Media Location & Search Engine Analytics

Location Analytics - Sources of Location Data, Categories of Location Analytics, Location Analytics and Privacy Concerns, Location Analytics Tools

Search Engine Analytics - Types of Search Engines, Search Engine Analytics, Search Engine Analytics Tools

Chapter 5

Social Information Filtering

Social Information Filtering - Social Sharing and filtering , Automated Recommendation systems, Traditional Vs social Recommendation Systems
Understanding Social Media and Business Alignment,
Social Media KPI, Formulating a Social Media Strategy,
Managing Social Media Risks

Chapter 6

Social Media Analytics Applications and Privacy

Social media in public sector - Analyzing public sector social media, analyzing individual users, **case study.**

Business use of Social Media - Measuring success, Interaction and monitoring, case study.

Privacy - Privacy policies, data ownership and maintaining privacy online.

Text Books

<u>Seven Layers of Social Media Analytics_ Mining Business Insights from Social Media Text, Actions, Networks, Hyperlinks, Apps, Search Engine, and Location Data,</u> Gohar F. Khan,(ISBN-10: 1507823207).	
Analyzing the Social Web 1st Edition by <u>Jennifer Golbeck</u>	
<u>Mining the Social Web_ Analyzing Data from Facebook, Twitter, LinkedIn, and Other Social Media Sites,</u> Matthew A Russell, O'Reilly	
<u>Social Media Analytics [2015],</u> Techniques and Insights for Extracting Business Value Out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press	
Charu Aggarwal (ed.), Social Network Data Analytics, Springer, 2011	

References:

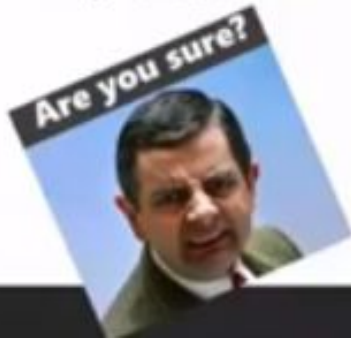
1.	<u>Social Media Analytics [2015]</u> , Techniques and Insights for Extracting Business Value Out of Social Media, Matthew Ganis, Avinash Kohirkar, IBM Press
2.	<u>Social Media Analytics Strategy_ Using Data to Optimize Business Performance</u> , Alex Gonçalves, APress Business Team
3.	<u>Social Media Data Mining and Analytics</u> , Szabo, G., G. Polatkan, O. Boykin & A. Chalkiopoulus (2019), Wiley, ISBN 978-1-118-82485-6

How many of you feel that you are part of a network?

- In fact, you are part of multiple networks!
- Do you feel any inertia/pull/attraction, being part of the networks?

No!

- Then what is this?



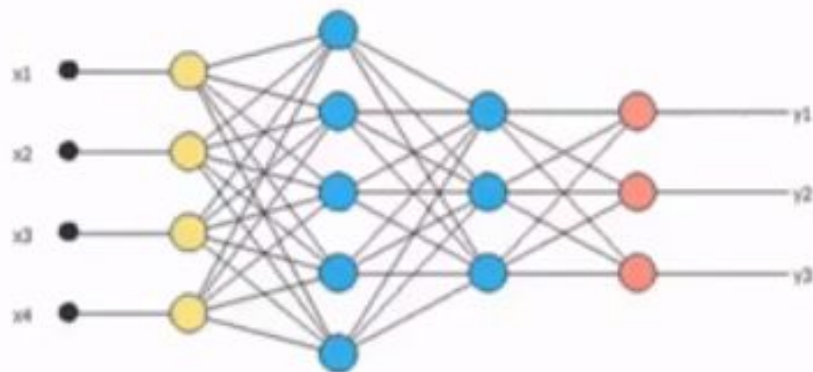
What is a Network?

It is not (only) a computer network.



What is a Network?

It is not (only) a neural network.



What is a Network?

A network is a simple (mathematical) abstraction of a complex system.

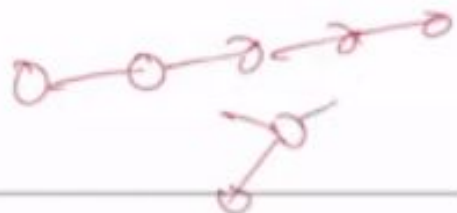
Networks are a general language for describing complex systems.

What is a Network?

"Network" is another nomenclature of "Graph"

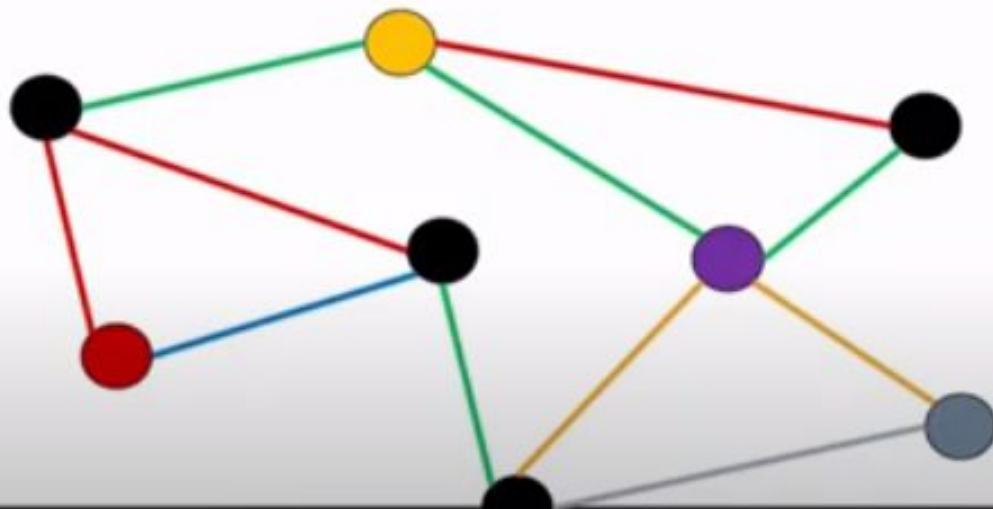
- aka "Complex Networks"

What is so
"complex" about it?



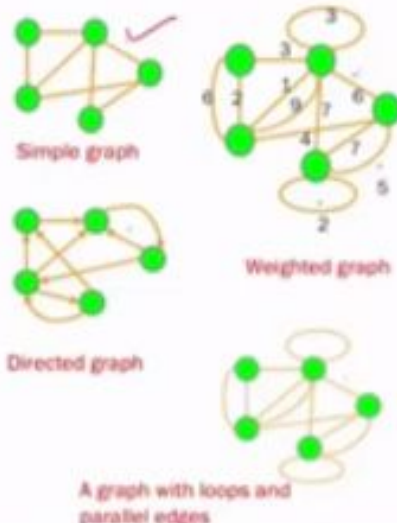
A complex network consists of nodes/vertices and edges/links, exhibiting non-trivial topological features that do not occur in a simple networks such as "lattices" or "random graphs", but often occur in graphs which model real systems.

Network = Individuals + connections

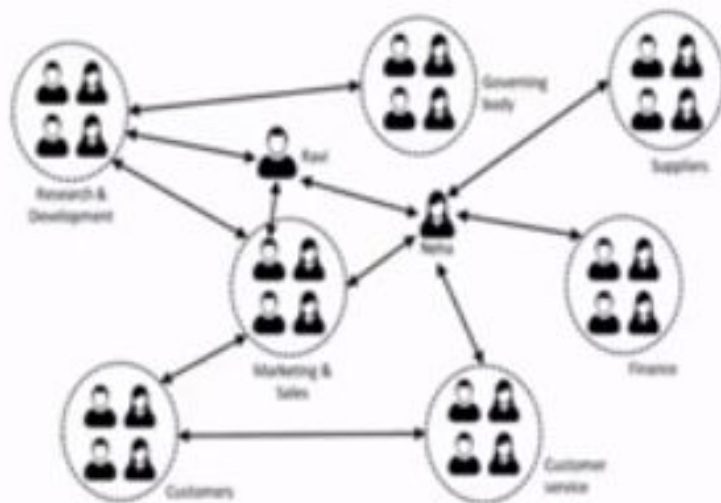


Network: Definition

- ❑ **Network (aka, graph):** An ordered pair $G(V, E)$, where V is a set of nodes/vertices, and E is a set of edges/links.
- ❑ Depending on the nature of application, the above definition may be revised or augmented, as follows:
 - ✓ **Undirected** (also called asymmetric, or irreversible), **directed** (also called symmetric, or reversible) edges, or **hyperedges**, etc.
 - ✓ Nodes and links with attributes/features like **weights**, **timestamps**, **textual features**, etc.
- ❑ **Self-loops:** Edges with same node as end nodes.
- ❑ **parallel edges:** More than one edge joining a pair of nodes.



What is Social Network Analysis?



Network

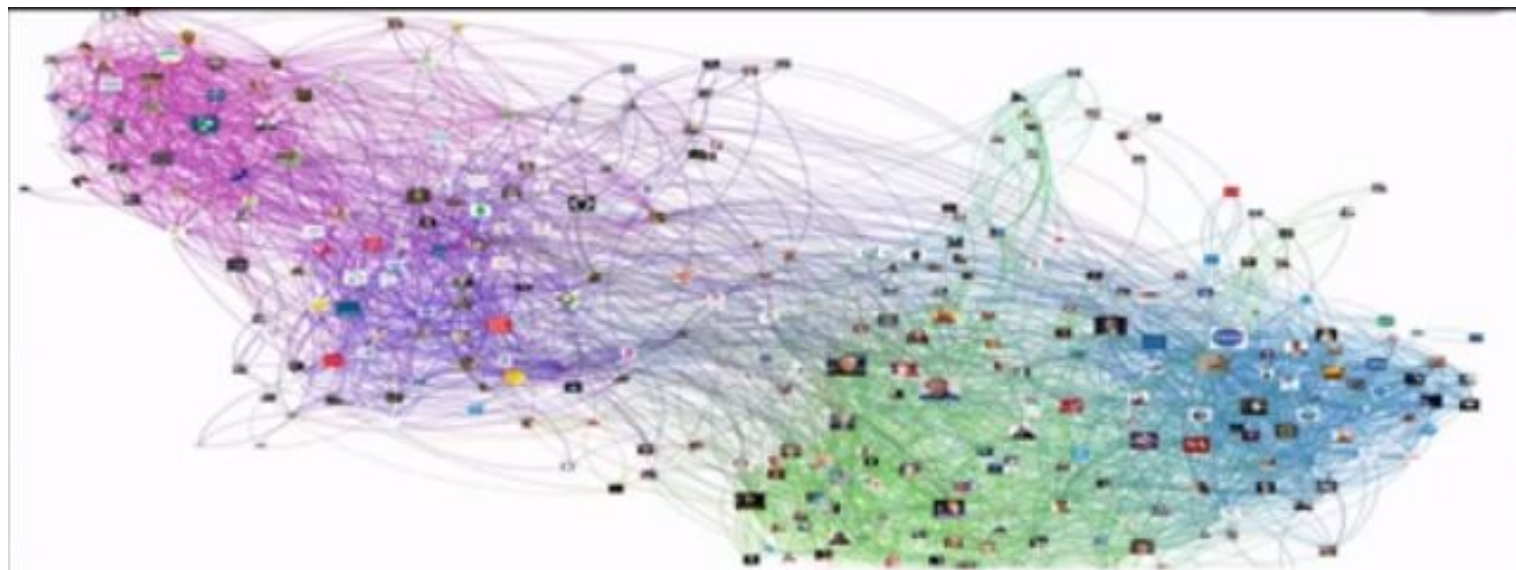
An abstract representation of relations among entities

Social Network

A simplified representation of the social structure characterized by actors and ties

Social Network Analysis

The application of networks and graph theory to analyze the relations present in a society



A sample of Twitter follower-followee network

(image source: <http://allthingsgraphed.com/2014/11/02/twitter-friends-network/>)

Social Network Analysis: Key Features

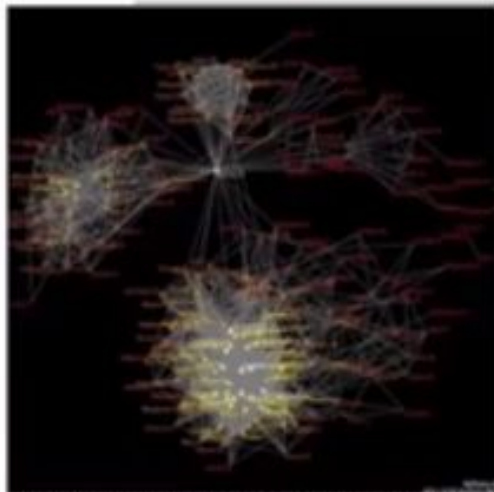
Required Knowledge Domains

- ☐ Sociology
- ☐ Psychology
- ☐ Mathematics
- ☐ Statistics
- ☐ Computer Science

Study Benefits

- ☐ To know the way social interactions influence a network
- ☐ To learn how the information flows inside a network
- ☐ To characterize roles of the individuals in a network
- ☐ To characterize communities inside a network
- ☐ To characterize the evolution of a network

SNA Applications: Social Media



An Example Map of a Facebook Friendship Network

<https://mathconceptions.wordpress.com/2012/01/16/application-snippet-friendship-and-influence-in-social-networks/>

- ☐ Identifying friendship structure
- ☐ Recommending friends, and groups, or pages to follow
- ☐ Identifying information propagation patterns
- ☐ Designing Humanitarian response applications after large-scale crises, etc.

SNA Applications: E-Commerce



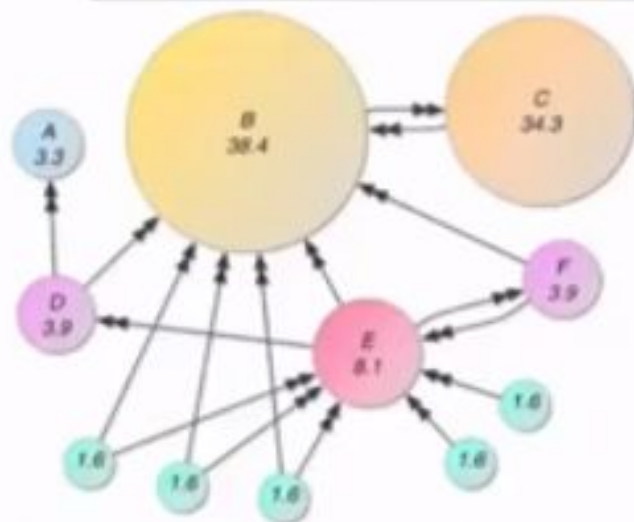
Influential Communities in Social Network

<https://towardsdatascience.com/influential-communities-in-social-network-simplified-fe5050d8e5a4>

- ❑ Customer profiling to Know the customers
- ❑ Product/Service recommendation based on customer profile
- ❑ Instigating viral marketing by pinpointing influential players
- ❑ "People like you buy", "Frequently bought with this", or "Frequently browsed", "Trending" are common buzzwords



SNA Applications: Web Search Optimization



Toy Example of Google's PageRank
<https://en.wikipedia.org/wiki/PageRank>

- ☐ Ranking the webpages based on hyperlinks
- ☐ Ordering of displayed pages based on ranks
- ☐ Displaying webpages based on user search profile



bing

YAHOO!



Aol.

Baidu 百度



SNA Applications: Cybercrimes & Terrorism



9/11 Terrorist Network

<http://www.orgnet.com/hijackers.html>

- Online fraud
 - Fake news propagation
 - Cyber bullying/trolling,
 - Sharing pornographic materials
-
- Terrorists often use social media to communicate as well as to brainwash innocent people

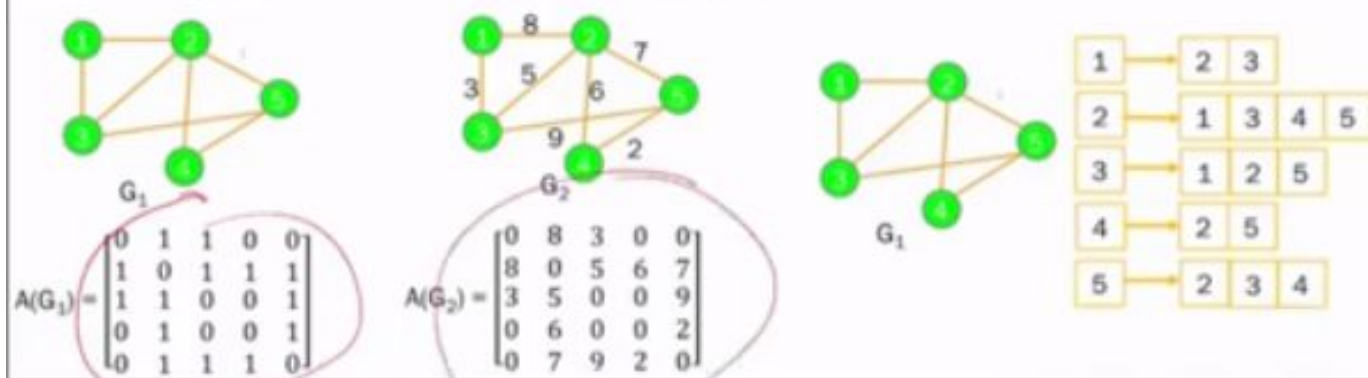
SNA Applications: Scientific Research & Academic Collaboration



- ☐ Scientific authors cite (refer to) the works of other authors
- ☐ Various networks of scholarly articles may be formed exploiting this relationship
 - ☐ Paper-paper citation network
 - ☐ Paper-paper co-citation network, etc.
- ☐ Various scholar networks may be formed using these relationships
 - ☐ Author collaboration network
 - ☐ Author citation network
 - ☐ Author co-citation network, etc.

Network Representation: Adjacency Matrix/List

An adjacency matrix $A = (a_{ij})$ for a graph $G(V, E)$ is a square matrix of dimension $|V| \times |V|$ such that each element a_{ij} of A indicates the existence of an edge between the node v_i and node v_j (also the weight of the corresponding edge in case of a weighted graph) in G .



Network Types: **Link-centric View**

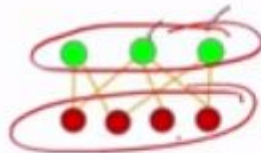
Unipartite Network



- ❑ Consists of a vertex set V and an edge set E . There is no restriction on the formation of edges between nodes of the network
- ❑ Example: An organizational LAN, where nodes are the devices, and edges are the local area links.
- ❑ Used to model the situation when links can join any pair of nodes of the network

Network Types: **Link-centric View**

Bipartite Network



- ❑ Consists of a vertex set V that is divided into two sets V_1 and V_2 that are disjoint and independent. Each edge of the network connects a vertex in V_1 to another vertex in V_2
- ❑ Example: An e-commerce user-product network. One part consists of the users, the other part consists of the products, the links are based on the basis of who bought what.
- ❑ Generalization of Bipartite network is n-partite networks, where the vertex set is partitioned into n number of part, an edges join a node from one part with a node from another part.

Network Types: **Link-centric View**

Signed Networks

- Consists of a set of nodes V , a set of edges E , and a function $f: E \rightarrow \{+, -\}$ that assigns each edge a positive or a negative sign

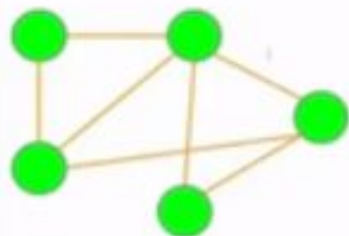


- Example: Consider a social media website that allows users to tag other users as friends or foes. The positive edges are friendship links and negative links are between foes.



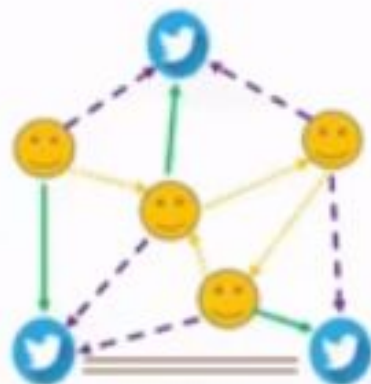
- studied specifically in the context of balance and status theory which determines the stability or existence of certain types of structural patterns in a network.

Network Types: **Node and Link-centric View:** Homogeneous Network



- Consists of a set of nodes V , all of which are of same type, and a set of edges E , all of which are of same type
- Example: Follower-Followee network of any kind.
 - ❖ Nodes are the users of the platform (all nodes are of same type),
 - ❖ Links are the follower-followee link between these nodes (all links are of same type)

Network Types: **Node and Link-centric View:** Heterogeneous Network



- Consists of a set of nodes V and a set of edges E , and two associated mapping functions, f_v and f_e , for nodes and edges, respectively. f_v maps a node to a node type and f_e maps an edge to an edge type
- Example: We consider a specific instance of Twitter network like the figure.
 - Two types of nodes:
 - Twitter Users
 - Tweet Posts
 - Four types of edges representing four types of relations between these nodes:
 - Posted-by (User – Post: Directed links)
 - Followed-by (User – User: Directed links)
 - Similar (Post – Post: Undirected links)
 - Retweet (User – Post: Directed links)

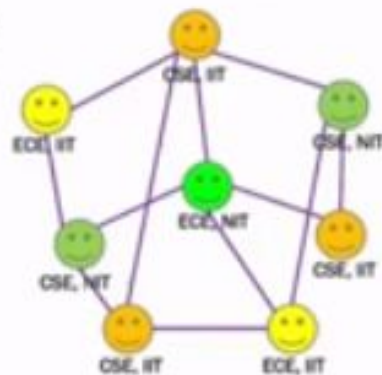
Network Types: **Node and Link-centric View:** Attributed Network

□ Consists of a set of nodes V and a set of edges E , and two associated mapping functions, f_v and f_e , for nodes and edges, respectively. f_v maps a node to a node attribute vector and f_e maps an edge to an edge attribute vector

□ Example: We consider a specific instance of Facebook network like the figure

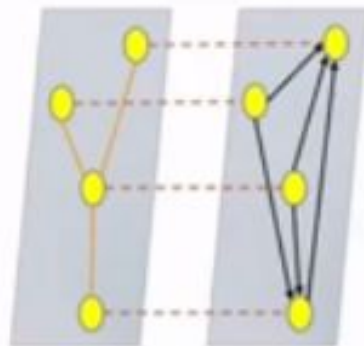
- Nodes are some Facebook users
- Edges are given by Facebook friendship relationship between these users
- Node attributes are the users' academic affiliations
- There is no edge attribute in this network

□ The example is a node-attributed network



Network Types: **Node and Link-centric View:** Multidimensional Network

- A special type of multilayer network where each layer represents a particular type of relationship among nodes
- Example 1: A special instance of Twitter network:
 - Nodes are Twitter users in both layer
 - Layer 1 edges: user – user similarity links (based on mutual interests) – Undirected links
 - Layer 2 edges: user – user follower-followee links – Directed links
- Note: In this example, each layer is node homogeneous.

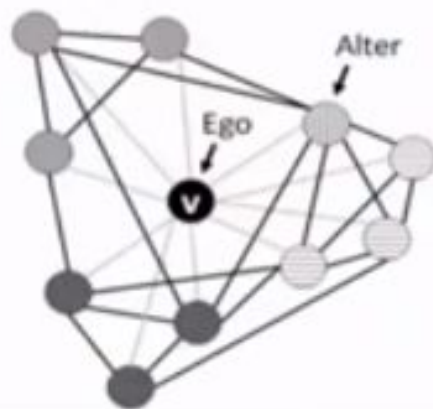


Network Types: **Local View:**

Ego-centric Network

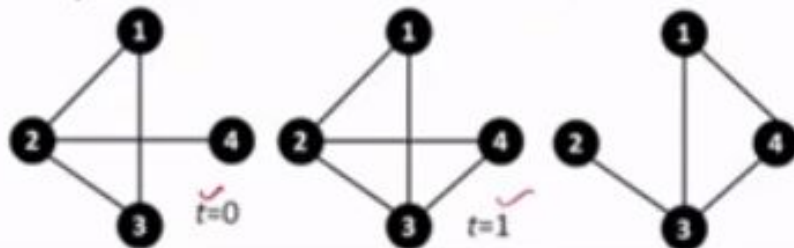
- A network of the form $G(V, E, u)$, that corresponds to a node $u \in V$, usually known as the 'ego', and consists of the node u as the central node, the nodes that are connected directly to the node u , usually known as the 'alters', and the induced subgraph for the same.

- Example: A subgraph of a Facebook Friendship Network:
 - Ego node corresponds to a user,
 - Alter nodes are his Facebook friends from different capacities and affiliations.



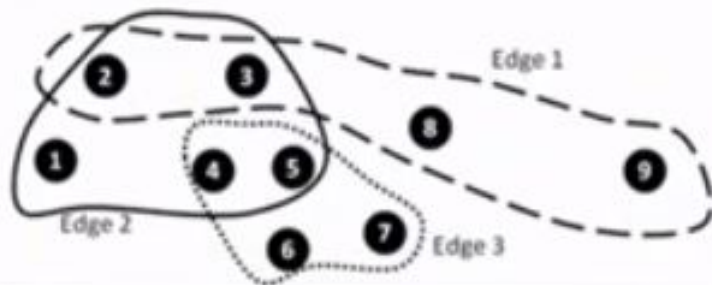
Network Types: **Temporal View:** Time-varying Network

- Consists of a set of nodes V and a set of edges E where each edge $e_{ij} \in E$ is represented by a three- tuple $e_{ij} = (v_i, v_j, t_{ij})$. Here, v_i and v_j are two end-points, and t_{ij} indicates the persistence duration of the edge e_{ij}
- Example: Person-to-person communication network over a span of time. The visible components are snapshots of the network at different time instances.



Network Types: **Generalized View:** Hypergraph

- Defined by a set of nodes V and a set of edge or hyperedges E , where each hyperedge e connects multiple nodes of the hypergraph
- Example: A special representation of Coauthorship Network:
 - ◆ Nodes are authors
 - ◆ Papers are hyperedges connecting the coauthors of the paper



Popular Real-world Networks

☐ Social Network

- Telephone call network
- Email message network
- Film actor collaboration network
- Academic co-authorship network

☐ Biological Network

- Protein-protein interaction networks
- Genetic regulatory networks
- Neural networks
- Metabolic networks
- Food Web
- Cell signalling networks

☐ Information Network

- World Wide Web (WWW)
- Citation network

☐ Technological Network

- Electric power grids
- Networks of airline routes
- Network of Railway Routes
- Electronic circuits
- Delivery networks of post-office/Courier
- The Internet

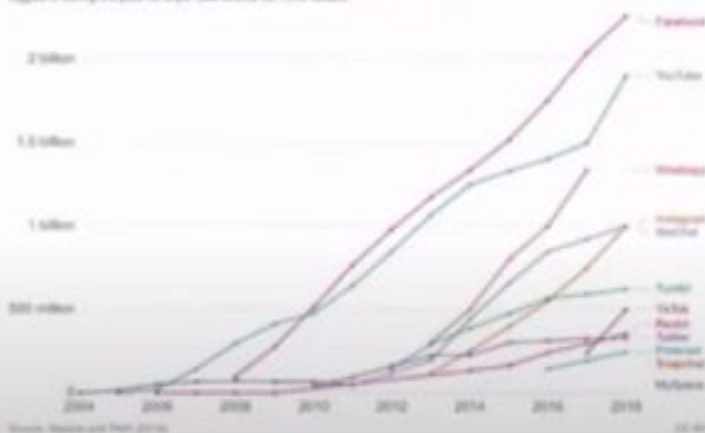
☐ Language Network

- Network formed by using the persons speaking a particular language

The Rise of Online Social Media: Some Interesting Facts

Number of people using social media platforms, 2004 to 2018

Estimates correspond to monthly active users (MAUs). Facebook, for example, measures MAUs as users that have logged in during the past 30 days. See source for more details.



❑ Online Social Media set in motion in early 2000

❑ First social networking and video website, **Bolt**, launched in 1996

❑ First social media site to reach a million monthly active users, **MySpace** in 2004

❑ 3.81 billion active social media users 2020

❑ 9.2% year-on-year increase from 3.48 billion in 2019

❑ 49.03% of the world's population (7.77 billion) is active on social media in 2020

<https://www.statista.com/chart/10000/number-of-social-media-users-in-the-world-2004-2018>

<https://www.statista.com/chart/10000/number-of-social-media-users-in-the-world-2004-2018>

Applications of network analysis

Node classification/coloring

Link prediction (friend/ad./product recommendation)

Growth and virality prediction

Network-centric/personalized community detection

Sampling and summarization

Information spreading and knowledge graph

Misinformation and anomaly detection

..... (any many more)

Ch 1. SOCIAL MEDIA ANALYTICS: AN OVERVIEW

Contents

- Types of Social Media
- Characteristics of Social Media
- Purpose of Social Media Analytics Social Media vs. Traditional Business Analytics
- Seven Layers of Social Media Analytics
 - Layer One: Text
 - Layer Two: Networks
 - Layer Three: Actions
 - Layer Four: Mobile
 - Layer Five: Hyperlinks
 - Layer Six: Location
 - Layer Seven: Search Engines
- Types of Social Media Analytics
 - Descriptive Analytics
 - Predictive Analytics
 - Prescriptive Analytics
- Social Media Analytics Cycle
- Challenges to Social Media Analytics

Types Of Social Media :

- Based on **Authentication or access mechanisms**, social media tools are available in two forms:
 - Internet-based
 - Smartphone-based.
- **Internet-based** social media platforms are generally accessed through e-mail IDs.
Example : Facebook, LinkedIn, Cyworld, and Google+ are examples of Internet-based social media.
- **Smartphone-based** social media platforms are accessed through mobile phone numbers; that is, users can only log in using mobile phone numbers.
Example : KaKao Talk, Tender, and 1KM
- Internet-based social media platform can also be accessed through any device connected to the Internet, including a smartphone application but the authentication mechanism is still the same.

Characteristics of Social Media :

- **SOCIAL MEDIA IS MANY-TO-MANY**

- Social media enables interaction among the users in a many-to-many fashion. This is unlike conventional technological media such as print, radio, telephone, and television.

- **SOCIAL MEDIA IS PARTICIPATORY**

- Unlike conventional technologies, **social media encourages participation and feedback from users**. Social media users can participate in online discourse through blogging, comments, tagging, and sharing content.

- **SOCIAL MEDIA IS USER OWNED**

- While social media platforms are provided by corporations (such as Google and Facebook), **the content is generated, owned, and controlled by social media users**. Without the **user-generated contents and active involvement from the users, social media would be empty, non existing**

- **SOCIAL MEDIA IS CONVERSATIONAL**

- Because of the many-to-many abilities, **it allows the masses to communicate and collaborate in real time**.

Characteristics of Social Media :

- **SOCIAL MEDIA ENABLES OPENNESS**

- Social media provides new opportunities to access data and information through **Web 2.0 channels**.

- **SOCIAL MEDIA IS RELATIONSHIP ORIENTED**

- Most social media tools allow users to easily establish and maintain social and professional relationships and ties. **Some social media tools, such as Facebook, are solely focused on personal relationships, whereas others, such as Twitter, are focused on professional relationships.**

- **SOCIAL MEDIA IS FREE AND EASY TO USE**

- Being free and easy to use are two of the reasons that social media has expanded in such a rate.

Social Media Analytics: Overview

- Social media analytics is the process of **extracting valuable insights** from **social media data to perform decision making**.
- By analyzing social media data, **businesses can increase brand loyalty, generate leads, drive traffic, and make forecasts**.
- Social media analytics can also be used to increase awareness of a brand and drive users to a website for the latest news and information.
- Social media growth, tools, and big data also present **huge opportunity to market products, enhance brand loyalty, network with customers, crowd-source ideas, drive sales, and mine business insights**.

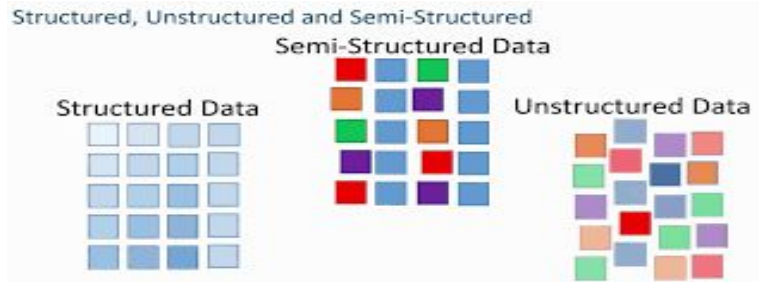
Purpose of Social Media Analytics

- The purpose of social media analytics is to enable **informed decision making** by leveraging social media data. This involves answering questions such as:
 - What are customers saying about a brand or product on social media?
 - Which content posted on social media is resonating more with customers?
 - How can social media data be used to improve products or services?
 - Is the social media conversation about a company, product, or service positive, negative, or neutral?
 - How can social media be used to promote brand awareness?
 - Who are the influential followers, fans, and friends on social media?
 - Who are the influential nodes (people and organizations) on social media and their position in the network?
 - Which social media platforms are driving the most traffic to a corporate website?
 - Where are the social media customers located geographically?
 - Which keywords and terms are trending on social media?
 - How active is social media in a business and how many people are connected with the company?
 - How are competitors performing on social media?

Social Media Vs Traditional Business Analytics

- The main difference between social media analytics and traditional business analytics is the **source, type, and nature of the data** being mined.
- Social media data is **diverse, high volume, real-time, and stored in third-party databases** in semistructured and unstructured formats, while traditional business data is mostly stored in **databases and spreadsheets in machine-readable i.e. Structured** format.
- Social media data is **socialized in nature and originates from the public internet**, while traditional business data is **bureaucratic and formal in nature and is controlled by organizations**.
- **The value of** social media data is determined by the **extent to which it is shared with other social entities**, while the value of traditional business data is often **confined within organizational databases and serves as a source of competitive advantage**.

Types of Data :



Unstructured Data

The university has 5600 students. Shaun (ID Number: 160801), 18 years old Communication study. Linh with ID number 160802, majoring in Accounting and is 20 years old. Ahmed from Psychology study program, 19 years old, ID number 160803.

Semi-Structured Data

```
<University>
  <ID Number="160801">
    <Name="Shaun">
    <Age="18">
    <Program="Communication">
  <ID Number="160802">
    <Name="Linh">
    <Age="20">
    <Program="Accounting">
  ..... </University>
```

Structured Data

ID	Name	Age	Program
160801	Shaun	18	Communication
160802	Linh	20	Accounting
160803	Ahmed	19	Psychology

Social Media Vs Traditional Business Analytics

Social Media Analytics	Business Analytics
Semistructured and unstructured data	Structured data
Data is not analytical friendly	Data is analytical friendly
Real-time data	Mostly historical data
Public data	Private data
Stored in third-party databases	Stored in business-owned databases
Boundary-less data (i.e., Boundary within the Internet)	Bound within the business intranet

Social Media Vs Traditional Business Analytics

Social Media Analytics	Business Analytics
Data is high volume	Data is medium to high volume
Highly diverse data	Uniform data
Data is widely shared over the Internet	Data is only shared within organizations
More sharing creates greater value/impact	Less sharing creates more value
No business control over data	Tightly controlled by business
Socialized data	Bureaucratic data
Data is informal in nature	Data is formal in nature

Seven Layers of Social Media Analytics

- Social media has a **minimum seven layers of data**
- Each layer carries potentially valuable information and insights that can be harvested for business intelligence purposes.
- Out of the seven layers, some are visible or easily identifiable (e.g., text and actions) and other are invisible (e.g., social media and hyperlink networks).
- The following are seven social media layers
 1. Text
 2. Networks
 3. Actions
 4. Hyperlinks
 5. Mobile
 6. Location
 7. Search engines

Seven Layers of Social Media Analytics



All Are
Interconnected !!!!

Figure 2. Seven layers of social media analytics

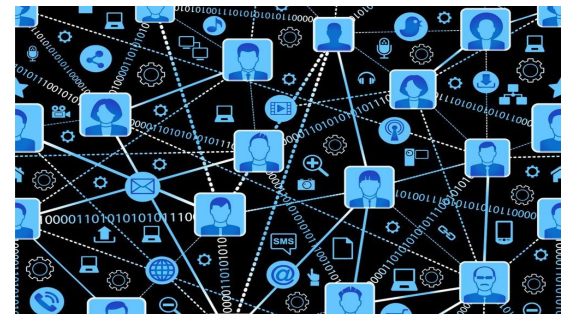
1. Text :

- TEXT is one of the fundamental elements of the social media platforms.



- Textual elements of social media **include comments, tweets, blog posts, status updates.**
- Social media text analytics, also known as **text mining**, is a **technique to extract, analyze, and interpret hidden business insights from textual elements of social media content.**
- Organizations use **text analysis techniques to extract hidden valuable meaning, patterns, and structures from the user-generated social media text for business intelligence purposes.**
- It is used to understand user **sentiments and identify emerging themes and topics.**
- **For example**, Text Analytics is useful in gaining a **quick and accurate understanding of the emotion and sentiment expressed over social media channels (e.g., tweets or Facebook comments) related to a brand or a new product launch.**

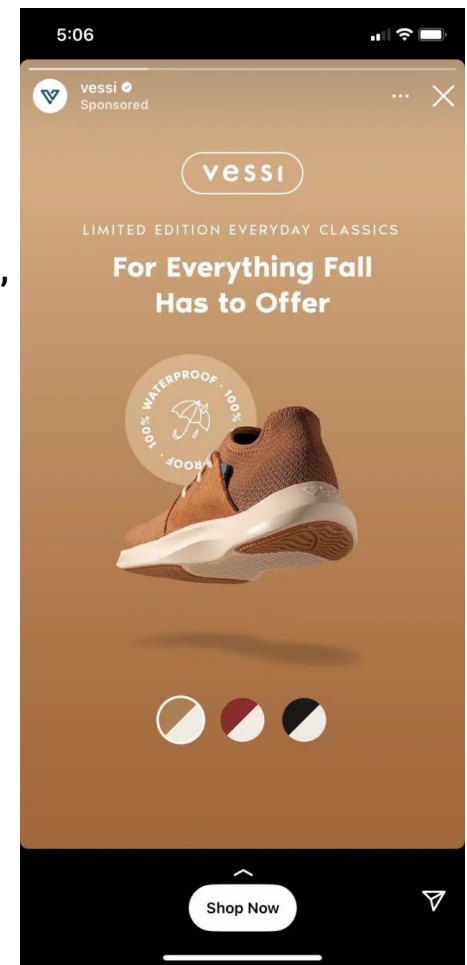
2. Network :



- This includes the **connections between users** and the **relationships between them**, such as **followers and friends on social media**. It is used to **identify influential nodes (people and organizations)** and their position in the network.
- A social network is a group of nodes and links
 - nodes can represent social entities such as **people and organizations**.
 - **Links represent their relationships, such as friendship and trade relations.**
- Twitter follow-following network is an example of an online social media network. **In a Twitter follow-following network, nodes are the Twitter users, and links among the nodes represents the follow-following relationship (i.e., who is following whom) among the users.**
- When applied to **Computer Supported Collaborative Learning CSCL**, **SNA(Social Network Analysis)** is used to help understand how learners collaborate in terms of amount, frequency, and length, as well as the quality, topic, and strategies of communication

3. Actions :

1. **Actions:** This includes the **actions taken by users on social media, such as likes, comments, shares, views, clicks and other interactions.**
2. It is used to measure **popularity, influence, and prediction** in social media.



3. Actions :

- Social media **actions analytics** deals with extraction, analysis, and interpretation of the insights contained in the actions performed by social media users.
- Social media **actions** are of great value to social media marketers because of their role in increasing revenue, brand value, and loyalty.
- Organizations can **employ actions analytics** to measure popularity and influence of a product, service, or idea over social media.
 - For example, a brand marketer might be interested to know how popular their new product is among social media users. Analyzing the Facebook likes and Twitter mentions, for example, can provide an answer to the question.

4. Mobile :

- Mobile analytics deals with measuring and optimizing user engagement with mobile applications .
- This includes data related to the **use of social media on mobile devices, such as the type of device used, the operating system, and the location of the user**
- **MOBILE WEB ANALYTICS** Mobile web analytics is mostly **focused on characteristics, actions, and behaviors of mobile website visitors; that is, the visitors to the mobile version of your company's website.**
- **APPS ANALYTICS** Apps analytics, however, **deal with understanding and analyzing mobile application users' characteristics, actions, and behaviors.**

5. Hyperlinks :

- Hyperlinks are **the pathways of social media traffic.**
- Hyperlinks are **references to web resources (such as a website, document, and files) that users can access by clicking on it.**
- **Hyperlinks can link resources within a document (inter-linking) and among documents (intralinking). For example, clicking on a hyperlink in a tweet can link you to other resources (e.g., websites) available over the Internet.**
- This includes the **links between social media posts and other websites**, as well as the content of those linked websites. **It is used to reveal Internet traffic patterns and sources of incoming or outgoing traffic.**
- Hyperlink analytics can also reveal, for example, **Internet traffic patterns and sources of the incoming or outgoing traffic to and from a website**

6. Location :

- This includes data on the **geographical location of users** and the **location-specific content** they engage with on social media.
- It is used to mine and map the locations of social media users, content, and data.
- Applications :
 - Geo Map related apps
 - Creation of bus services like CitiFlo
 - Fitness related apps

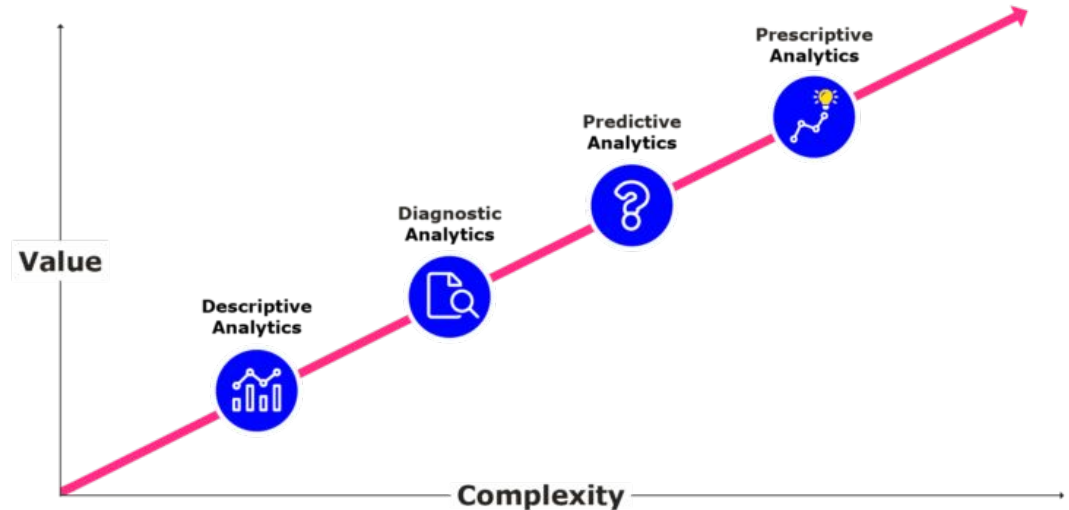
7. Search Engines :

- This includes data on **how users find and access social media content** through **search engines**, as well as the keywords they use in their searches.
- It is used to **analyze historical search data for trends analysis, keyword monitoring, and advertisement history.**

Social Media Analytics Types

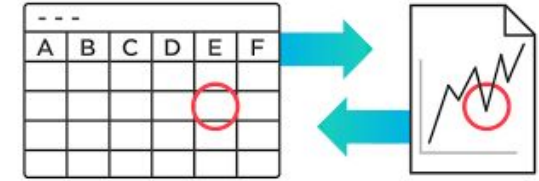
Like any business analytics, social media analytics can take four forms:

- 1) Descriptive Analytics (What)
- 2) Diagnostic Analysis (Why)
- 3) Predictive analytics
- 4) Prescriptive analytics (How)



1. Descriptive analytics

WHAT HAPPENED?



- Descriptive analytics is the **process of parsing historical data to better understand the changes that have occurred in any situation**
- Descriptive analytics can **help to identify the areas of strength and weakness**
- This involves gathering and describing social media data in the form of **reports, visualizations, and clustering to understand a business problem**. Examples include actions analytics (e.g., number of likes, tweets, and views) and text analytics.
- Descriptive analytics answer the question “**what is happening**”? “**What happened**”? **Comments and posts can be grouped together, for instance, for the purpose of sentiment analysis**
- Any time you gather a **lot of similar data points in order to analyze them looking for patterns, sentiments, and/or trends, you’re dealing with descriptive analytics.**

2. Diagnostic analytics

- Diagnostic SMA analytics **looks into the questions of “why something happened?”**
- Diagnostic Analytics, as the name suggests, **gives a diagnosis to a problem. It gives a detailed and in-depth insight into the root cause of a problem.**
- Enablers of diagnostics analytics include **inferential statistics, behavioural analytics, correlations & retrospective analysis and outcome being cause and effect analysis of a business issues.**

2. Diagnostic analytics

- An **example for diagnostic analytics** can be an **e-commerce** company. Given the situation that the sales of the company have gone down.
- The possible reasons behind this problem can be: the form didn't load correctly, the shipping charges are high, and not enough payment methods are available.
- Taking the help of diagnostic analytics, **the company comes out with a specific reason and then works on that to resolve the issue.**

3. Predictive analytics :

- **Predictive analytics:** This involves analyzing large amounts of accumulated social media data to **predict a future event.**
 - For example, analyzing **social media posts to predict future purchasing behavior or using historical website visits to predict future sales figures.**
- Predictive analytics uses models like **data mining, AI, and machine learning to analyze current data and forecast what might happen in specific scenarios.**

3. Predictive analytics :

- It can range from simple things, like predicting **possible visits to a location based upon posts expressing that intention**, to forecasting entire trends and phenomena based upon mentions.
- Social listening tools **can help identify upcoming trends and shifts in consumer behaviour by analysing large volumes of social media data and indicating the shifting popularity of keywords**.

4. Prescriptive analytics :

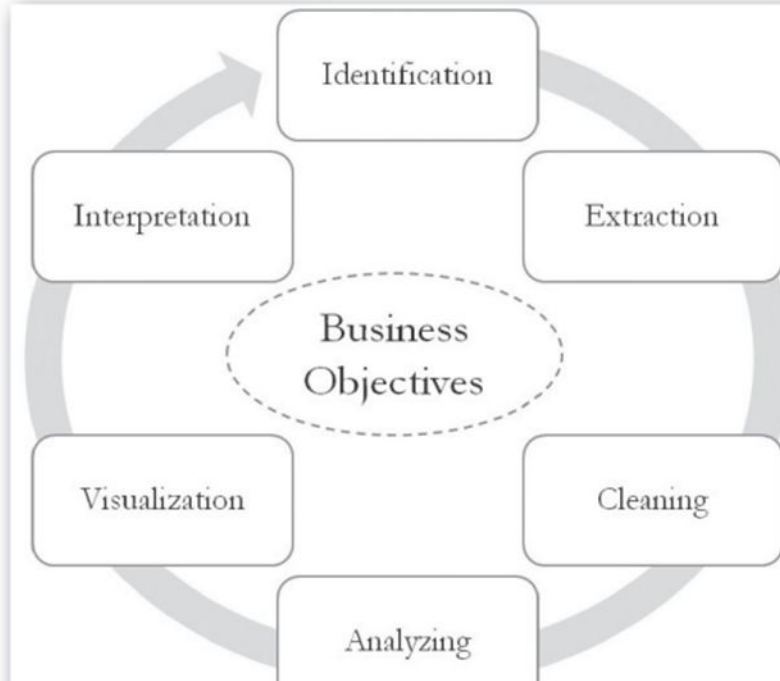
- Prescriptive analytics is a **combination of data and various business rules**. The data of **prescriptive analytics** can be both internal (organizational inputs) and external (social media insights).
- While **predictive analytics** help to predict the future, prescriptive analytics suggest the **best action to take** when handling a scenario
- When combined with **predictive analytics**, it adds the benefit of manipulating a future occurrence like mitigate future risk.

- **Prescriptive analytics:** This involves suggesting the best action to take when handling a scenario based on analyzing patterns of behavior. It has not yet been widely applied to social media data.
- Prescriptive analysis is **the analysis of data with the intention of providing the best way to proceed at any given moment.**
- This can be **applied to situations ranging from handling social media crises and incidents (“how well this type of change will be accepted by our target audience?”) to purchase preferences (“we’ve identified this group of customers – how do we optimize our sales process to their habits?”).**

Social Media Analytics Lifecycle

The social media analytics process involves six steps to mine desired business insights from social media data.

The process begins with defining business goals and objectives, and continues until these objectives are fully satisfied.



Step1 - Identification

- The identification stage of social media analytics involves **finding the right sources of data to analyze** in order to **gain valuable business insights**.
- The **data** should be aligned with the **business's objectives** and can come from **both official business-owned platforms, such as social media accounts and blogs, and nonofficial platforms such as Google search trends or Twitter search stream data**.
- It is important to **consider the business objectives when identifying the sources and types of data to be analyzed**.

Step2 - Extraction

- The extraction stage of social media analytics involves using **appropriate methods and tools to gather data from identified sources.**
- The **type (e.g., text, numerical, or network) and size of data** will **determine the method and tools suitable for extraction.**
- This **can include manual data collection for small-scale data and automated extraction using APIs (application programming interfaces) for larger data sets.**
- It is important to **consider privacy and ethical issues when mining data from social media platforms** and to have a **clear social media privacy policy in place to ensure that data handling and extraction practices do not violate user privacy.**
- Specialized tools may be needed to extract certain types of data, such as social network and hyperlink network data.

Step3 - Cleaning

- The cleaning step in social media analytics involves **removing unwanted data** from the collected data set.
- This can involve processes such as **cleaning, coding, filtering, clustering, and natural language processing** to remove irrelevant data.
- Both **automated and manual techniques** may be used for cleaning, depending on the type of data and the desired level of accuracy.

Step 4- Analyzing

- The analyzing stage of social media analytics involves using clean data to **identify valuable insights for the business.**
- The **approach and techniques used will depend on the type of data being analyzed and the tools and algorithms employed.**
- It is important **to maintain the integrity of the data** while **extracting meaningful insights** and to have a good understanding of the capabilities of the tools being used.

Step 5 - Visualization

- The visualization step in social media analytics involves **creating visual representations of the results of the analysis**
- Visualization can help **reveal hidden patterns, relationships, and trends in complex and large data sets**
- Different types of types of visualizations include **charts, graphs, trees, heat maps, 3D views**
- **Effective visualization** is important for **effectively communicating the results of the analysis to top management**
- **Text analytics** can result in a word cooccurrence cloud; **hyperlink analytics** will provide visual hyperlink networks; and **location analytics** can produce interactive maps.

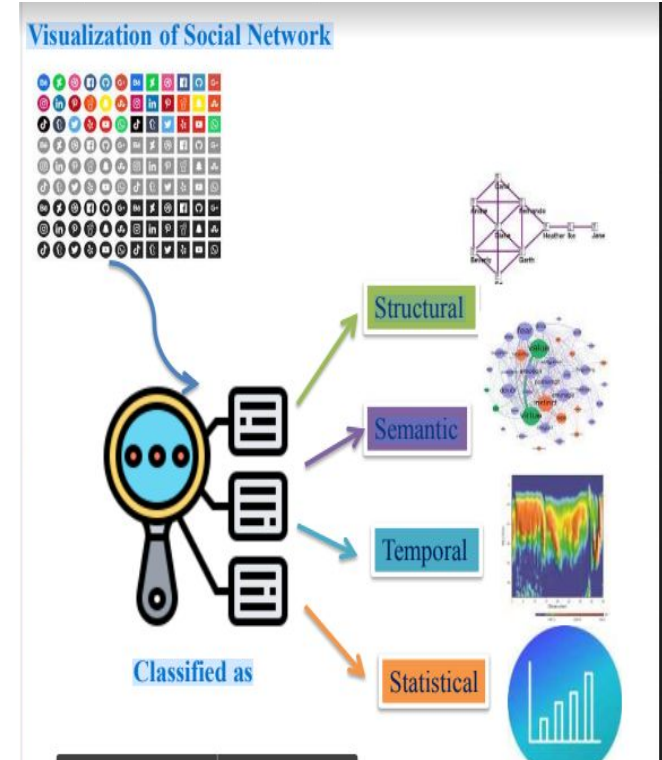
Depending on the type of data, different types of visualization are possible :

Network data (with whom)—network data visualizations can show who is connected to whom. For example, a Twitter following-following network chart can show who is following whom.

Topical data (what)—topical data visualization is mostly focused on what aspect of a phenomenon is under investigation. A text cloud generated from social media comments can show what topics/themes are occurring more frequently in the discussion.

Temporal data (when)—temporal data visualization slice and dice data with respect to a time horizon and can **reveal longitudinal trends, patterns, and relationships hidden in the data.**

Geospatial data (where)—geospatial data visualization is **used to map and locate data, people, and resources.**



Step 6 : Interpretation

- Interpret and translate analytics results into a **meaningful business problem**.
- Two strategies or approaches used are:
 - 1) Producing easily consumable analytical results and
 - 2) Improving analytics consumption capabilities

Challenges to Social Network Analysis

- **Volume and Velocity** is a challenge
 - Social media data **is large in size and is swiftly generated.**
 - **Capturing and analyzing millions of records that appear every second is a real challenge.** For example,
 - on Twitter, **three-hundred forty-two thousand tweets appear every minute, and on Facebook, one million likes are shared every twenty minutes.**
 - Capturing all this **information may not be feasible.**
 - **Knowing what to focus on is crucial for narrowing down the scope and size of the data.**
 - Luckily, sophisticated tools are being developed to handle high-volume and high-velocity data.

Challenges to Social Network Analysis

- **Diversity Challenge**- Social media users and content they generate are extremely diverse, multilingual, and vary across time and space.
- **Unstructuredness as a challenge**
 - Unlike the data stored in the corporate databases, which are mostly numbers, social media data is highly unstructured and consists of text, graphics, actions, and relations.
 - Short social media text, such as tweets and comments, has random grammatical structure, and is laden with abbreviations, acronyms, and emoticons (a symbol or combination of symbols used to convey emotional expressions in text messages), thus representing a great challenge for extracting business intelligence.

Social Media Analytics Tools

Layer of social media	Example of tools
Text	Discovertext Lexalytics Demo :Video Link Archivist Twitonomy Netlytic LIWC Voyant
Actions	Lithium Twitonomy Google Analytics Video Demo : Video Link SocialMediaMineR
Network	NodeXL UCINET Pajek Netminer Flocker Netlytic Reach Mentionmapp

Mobile	Countly
	Mixpanel
	Google Mobile Analytics
	Google Fusion Table
Location	Tweepsmap
	Trendsmap
	Followerwonk
	Esri Maps Agos
Hyperlinks	Webometrics Analyst VOSON
Research Engines	Google Trends

Questions

- Why it is important for business managers to understand and mine social media data?
- What is social media analytics, and how it is different from traditional business analytics?
- Briefly explain the seven layers of social media data. Support your answer with examples.
- Explain the social media analytics cycle.
- What ethical issues should be considered when mining social media data?
- What are some main challenges to social media analytics?
- Compare different social media analytics tools available in the market and explain their strengths and weakness.